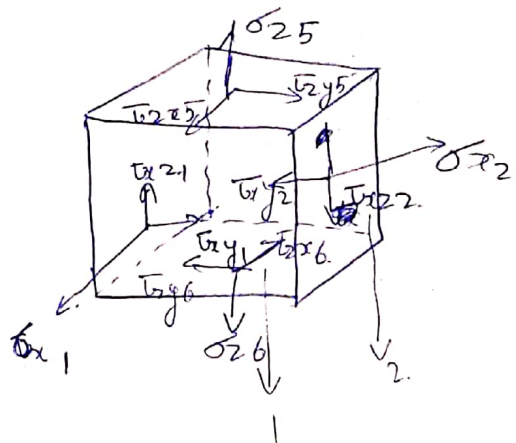
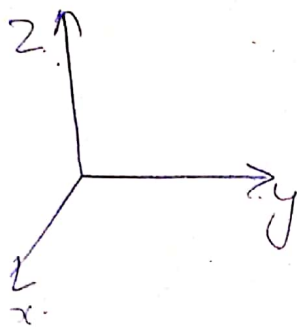


Differential eqs of equilibrium



x, y, z body forces in x, y, z dir.

in x direction

$$(\sigma_{x1} - \sigma_{x2}) dy dz + (\tau_{yx3} - \tau_{yx4}) dx dz + (\tau_{zx5} - \tau_{zx6}) dy dx + X dx dy dz = 0$$

dividing by $dx dy dz$.

Let $dx \rightarrow 0$
 $dy \rightarrow 0$
 $dz \rightarrow 0$

$$\frac{\sigma_{x1} - \sigma_{x2}}{dx} + \frac{\tau_{yx3} - \tau_{yx4}}{dy} + \frac{\tau_{zx5} - \tau_{zx6}}{dz} + X = 0$$

x dir $\rightarrow$

$$\frac{d\sigma_x}{dx} + \frac{d\tau_{yx}}{dy} + \frac{d\tau_{zx}}{dz} + X = 0$$

sim. ly.

y dir $\rightarrow$

$$\frac{d\tau_{xy}}{dx} + \frac{d\sigma_y}{dy} + \frac{d\tau_{zy}}{dz} + Y = 0$$

2d stress $\rightarrow$

$$\frac{\partial \bar{\sigma}_{xz}}{\partial x} + \frac{\partial \bar{\sigma}_{yz}}{\partial y} + \frac{\partial \bar{\sigma}_z}{\partial z} + \bar{z} = 0$$

$$\{\bar{T}_n\} = [\sigma] \{u\}$$



with 3 eq & 6 unknowns

3 displ $\rightarrow u, v, w$

6 stresses $\rightarrow \epsilon_x, \epsilon_y, \epsilon_z, \gamma_{xy}, \gamma_{yz}, \gamma_{xz}$

and

$$\epsilon_x = \frac{du}{dx}, \quad \epsilon_y = \frac{dv}{dy}, \quad \epsilon_z = \frac{dw}{dz}$$

$$\gamma_{xy} = \frac{du}{dy} + \frac{dv}{dx}, \quad \gamma_{yz} = \frac{dv}{dz} + \frac{dw}{dy}$$

$$\gamma_{xz} = \frac{du}{dz} + \frac{dw}{dx}$$

Compatibility eq.

→ from previous

$$\frac{\partial^2 \epsilon_x}{\partial y^2} + \frac{\partial^2 \epsilon_y}{\partial x^2} = \frac{\partial^2 \gamma_{xy}}{\partial x \partial y}$$

$\begin{matrix} x \\ \nearrow \\ z \\ \nwarrow \\ y \end{matrix}$

$$\frac{\partial^2 \epsilon_y}{\partial z^2} + \frac{\partial^2 \epsilon_z}{\partial y^2} = \frac{\partial^2 \gamma_{yz}}{\partial y \partial z}$$

$$\frac{\partial^2 \epsilon_z}{\partial x^2} + \frac{\partial^2 \epsilon_x}{\partial z^2} = \frac{\partial^2 \gamma_{zx}}{\partial z \partial x}$$

$$\frac{\partial^2 \epsilon_x}{\partial y \partial z} = \frac{\partial^3 u}{\partial x \partial y \partial z} \quad \text{---} \quad *$$

$$\frac{\partial \gamma_{yz}}{\partial x} = \frac{\partial^2 v}{\partial x \partial z} + \frac{\partial^2 w}{\partial x \partial y} \quad \text{---} \quad **$$

$$\frac{\partial \gamma_{zx}}{\partial y} = \frac{\partial^2 u}{\partial y \partial z} + \frac{\partial^2 w}{\partial x \partial y} \quad \text{---} \quad ***$$



$$\frac{\partial}{\partial x} [** + *** - *]$$

$$\frac{\partial^2 u}{\partial x \partial y \partial z}$$

$$2 \frac{d^2 \epsilon_x}{dy dz} = \frac{d}{dx} \left[-\frac{d \gamma_{yz}}{dx} + \frac{d \gamma_{xz}}{dy} + \frac{d \gamma_{xy}}{dz} \right]$$

Similarly

$$2 \frac{d^2 \epsilon_y}{dx dz} = \frac{d}{dy} \left[-\frac{d \gamma_{yz}}{dy} + \frac{d \gamma_{yz}}{dx} + \frac{d \gamma_{xy}}{dz} \right]$$

$$2 \frac{d^2 \epsilon_z}{dx dy} = \frac{d}{dz} \left[-\frac{d \gamma_{xy}}{dz} + \frac{d \gamma_{yz}}{dx} + \frac{d \gamma_{xz}}{dy} \right]$$

$$\frac{d \sigma_x}{dx} + \frac{d \tau_{xy}}{dy} + \frac{d \tau_{xz}}{dz} + X = 0$$

$$\sigma_x = \lambda e + 2\mu \epsilon_x$$

$$\tau_{xy} = \mu \gamma_{xy}$$

$$\tau_{xz} = \mu \gamma_{xz}$$

$$\lambda = \frac{\alpha \nu E}{(1+\nu)(1-2\nu)}$$

$$\mu = \frac{E}{2(1+\nu)}$$

$$\frac{d}{dx} \left[\lambda e + 2\eta \frac{du}{dx} \right] + \eta \frac{d}{dy} \left(\frac{du}{dy} + \frac{dv}{dx} \right) + \eta \frac{d}{dz} \left[\frac{du}{dz} + \frac{dw}{dx} \right]$$

$$+ X = 0$$

$$\lambda \frac{de}{dx} + 2\eta \frac{d^2 u}{dx^2} + \eta \left[\frac{d^2 u}{dy^2} + \frac{d^2 v}{dx dy} + \frac{d^2 u}{dz^2} + \frac{d^2 w}{dx dz} \right] + X = 0$$

$$\underbrace{\eta \frac{d}{dx} \left[\frac{du}{dx} + \frac{dv}{dy} + \frac{dw}{dz} \right]}_{e} + \underbrace{\eta \left[\frac{d^2 u}{dx^2} + \frac{d^2 u}{dy^2} + \frac{d^2 u}{dz^2} \right]}_{\nabla^2 u} + X = 0$$

$$(\lambda + \eta) \frac{de}{dx} + \eta \nabla^2 u + X = 0$$

for y

$$(\lambda + \eta) \frac{de}{dy} + \eta \nabla^2 v + Y = 0$$

for z

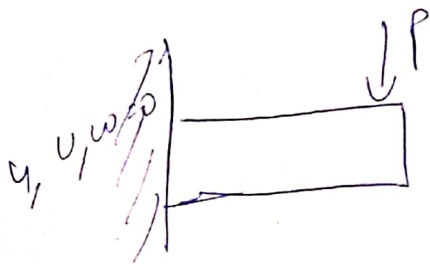
$$(\lambda + \eta) \frac{de}{dz} + \eta \nabla^2 w + Z = 0$$

$$\bar{X} = \sigma_x l + \tau_{xy} m + \tau_{xz} n$$

$$\bar{Y} = \tau_{xy} l + \sigma_y m + \tau_{yz} n$$

$$\bar{Z} = \tau_{xz} l + \tau_{yz} m + \sigma_z n$$

$$\begin{aligned} \bar{X} = & \left[\lambda e + 2\eta \frac{du}{dx} \right] l + \eta \left[\frac{du}{dy} + \frac{dv}{dx} \right] m \\ & + \eta \left[\frac{dw}{dx} + \frac{du}{dz} \right] n. \end{aligned}$$



$$\int \tau_{xy} dA = P$$